

HELPDESK - Planejamento Técnico e Modelo de Dados

Backend:

- Java 21
- Spring Boot 3
- Spring Security
- Spring Data JPA
- JWT
- Maven

Frontend:

- Angular 20+
- Angular Material
- RxJS
- JWT Interceptor

Banco de dados:

- PostgreSQL 16+

Arquitetura em Camadas

Controller



Service



Repository



Database

Exemplo:

TicketController



TicketService



TicketRepository



PostgreSQL

Estrutura Backend

src/main/java

**config/
security/
controllers/
services/
repositories/
entities/
dtos/
exceptions/
auditing/
utils/**

Estrutura Frontend

src/app

**core/
 auth/
 interceptors/
 guards/**

**shared/
 components/**

**modules/
 dashboard/
 users/
 tickets/
 comments/
 audit/**

layouts/

Segurança

Autenticação

JWT Bearer Token

Authorization: Bearer eyJhb...

Senhas

BCryptPasswordEncoder

Controle de Acesso

Spring Security com Roles:

ROLE_ADMIN
ROLE_SUPERVISOR
ROLE_TECHNICIAN
ROLE_REQUESTER

Exemplo:

```
@PreAuthorize("hasRole('ADMIN')")
```

Modelo de dados

Usuário:

Id	BIGSERIAL
Name	VARCHAR(150)
Email	VARCHAR(150)
Password	VARCHAR(255)
Created_at	TIMESTAMP
Updated_at	TIMESTAMP
Active	BOOLEAN
Role	INTEGER

Chamado:

Id	BIGSERIAL
Title	VARCHAR(200)
Description	TEXT
Status	INTEGER

Priority	INTEGER
Requester_Id	BIGSERIAL
Assigned_to_Id	BIGSERIAL
Created_at	TIMESTAMP
Updated_at	TIMESTAMP
Closed_at	TIMESTAMP

Comentários:

Id	BIGSERIAL
Ticket_Id	BIGSERIAL
User_Id	BIGSERIAL
Comment	TEXT
Created_At	TIMESTAMP

Auditoria:

Id	BIGSERIAL
User_Id	BIGSERIAL
Action	VARCHAR(100)
Entity_Type	INTEGER
Entity_Id	BIGSERIAL
Old_Value	JSONB
New_Value	JSONB
Created_At	TIMESTAMP